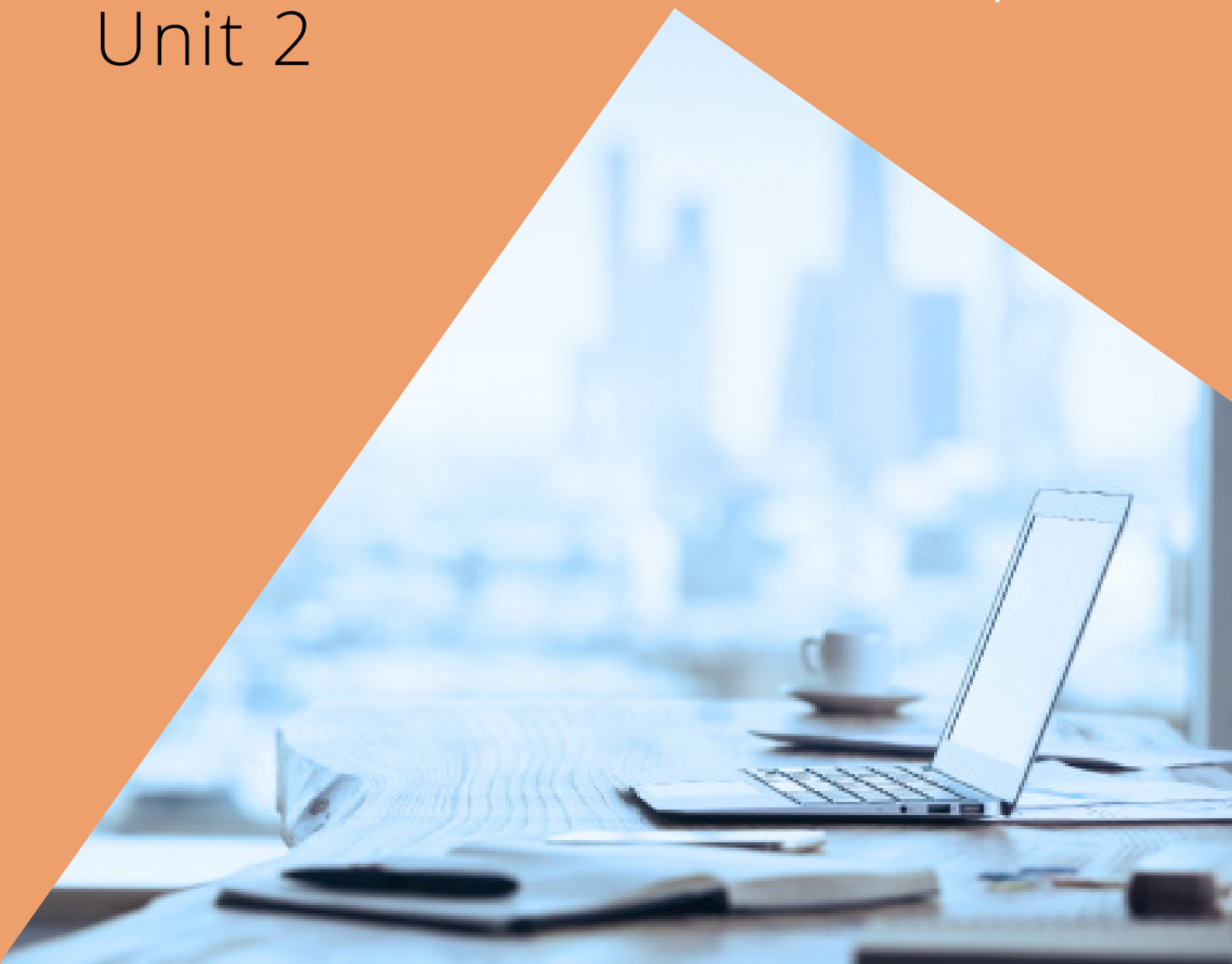


vision**2**learn

Understanding Coding Level 2

Hints
and
Tips

Unit 2



Contents

Section 1 – Understand basic computer terminology	4
Section 2 – Understand key principles when writing code	5
Section 3 – Understand coding acronyms and terminology	7

Hints and Tips Unit 2

The information in this guide will be helpful to you when you are ready to complete your assessment for the following unit:

Unit 2: Understanding terminology used in coding
Level 2

Section 1 – Understand basic computer terminology

- Q1.** In the table below **define** what the Central Processing Unit (CPU) does within a computer and, **explain** how Random-Access memory is used within a computer.

This question covers both learning outcomes 1.1 and 1.2. For this you simply need to define what the CPU and RAM do. What processes does each of these carry out? What are the different roles that they carry out.

- Q2.** In the box below, using an example, **define** the term 'data'.

For this question you simply need to think of an example of what data is in relation to computing.

- Q3.** In the table below, **list** the **key purpose** of five common data types used in coding. You may find it helpful to use examples.

This question is asking you to consider what the purpose of the five common data types is. The first one is already provided for guidance so all you need to do for that one is consider what its key purpose is and then add the other four. As mentioned in the question it may be helpful to give examples of what each data type is used for.

- Q4.** In the box below **identify** how a computer stores data and, identify what other items/devices are used to store/backup data.

This is an identify question so you can list your responses. Think about how your own computer stores data and the different devices that are commonly used to store data. You may also want to consider older types of storage that may not be common now but are still used by certain organisations.

Section 2 – Understand key principles when writing code

- Q1.** In the table below, using the above JavaScript example, **explain** how the variables have been used and, **define** what an operand is and an operation.

This is an explain question so try to provide good detail in your response. You should look at the code example at the top of section 2 to help you with this. You then need to also give a definition of what an Operand and Operation are also using the code example to help you answer this part of the question.

- Q2.** In the box below, using the above JavaScript example, **describe** the principles of functions.

As with question 1, you should use the code example at the top of session 2 as well as the learning materials and describe what the principles of functions are. What are they doing, what is their purpose?

- Q3.** In the box below, using the above JavaScript example, **define** what is meant by iterations.

For this question you simply need to define what iterations are in relation to the coding example. What part of the code is an iteration, what is the purpose if an iteration?

- Q4.** In the box below, using the above JavaScript example, **explain** what is meant by control flow.

This is an explain questions so try to provided good detail in your response. What does control flow do in relation to coding? How does this relate to the code example provided at the top of session 2?

Q5. In the box below, **explain** the use of queues, stacks, and lists.

This is an explain question so try to provide good detail in your response. What are each of these used for, what is their purpose?

Q6. In the box below, **explain** what is meant by the term 'collection'.

This is an explain question so try and provided good detail in your answer. Think about what a collection is in terms of coding, what is it used for?

Q7. In the box below, using the above Python example, **define** what is meant by recursions.

For this question you need to consider what a recursion is. You should use your learning from the course materials and also the code example at the top of section 2 to help you with this. What is the purpose of recursions? Do they occur in the code example?

Section 3 – Understand coding acronyms and terminology

Q1. In the box below **describe** what is meant by a 'unit of code'.

This is a describe question so try to provide good detail in your answer. What is a unit of code? It may help to give examples.

Q2. In the table below, **list** the characteristics of DRY and WET code, and **explain** the differences between the two. (Look at the examples of DRY and WET code in Session 3 on page 1 to help you answer this question.)

Using your learning from the course materials make a list of the characteristics of DRY and WET code then in the last part of the question explain the differences between the two. You may want to give examples of when these are used.

Q3. Do a little research to find examples of programs that have been written in open source code and closed source code. Use this research to help you to **describe** what is meant by 'open sourced' code, and what is meant by 'closed sourced' code in the box below. (Make sure you evidence websites and/or books used).

This is a describe question so try and provide good detail in your response. As suggested do some research to assist you. Ensure that any information that you include in your response is referenced.